1INF03 - Análisis de datos
2026-2

SEMANA 1: INTRODUCCIÓN A KDD

"KNOWLEDGE DISCOVERY IN DATABASES"

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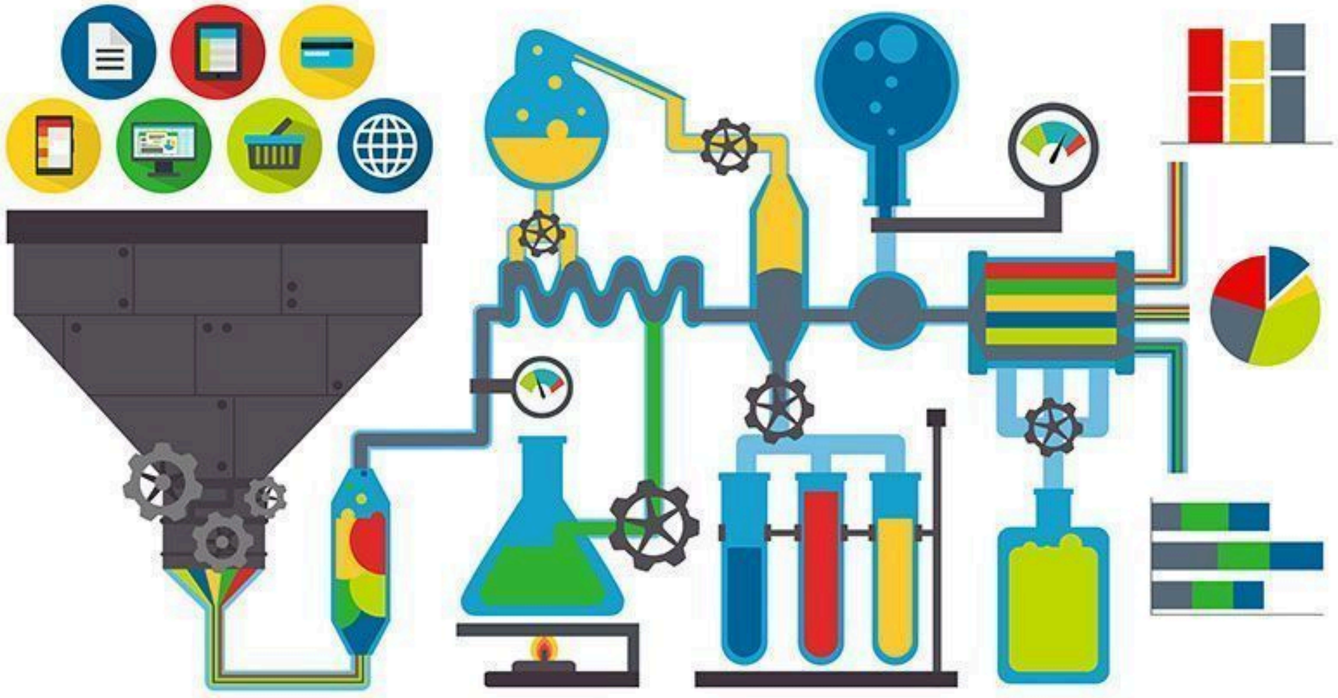


Imagen: http://www.deepphat.co.uk/2018/04/data-miner.html

- DATO -
- INFORMACIÓN -
- CONOCIMIENTO -

¿QUÉ ES UN DATO?

- Representación simbólica de un atributo o variable
- No contiene información en sí mismo
- Aplicando una hipótesis, se transforma en información
- El procesamiento de los datos proporciona información

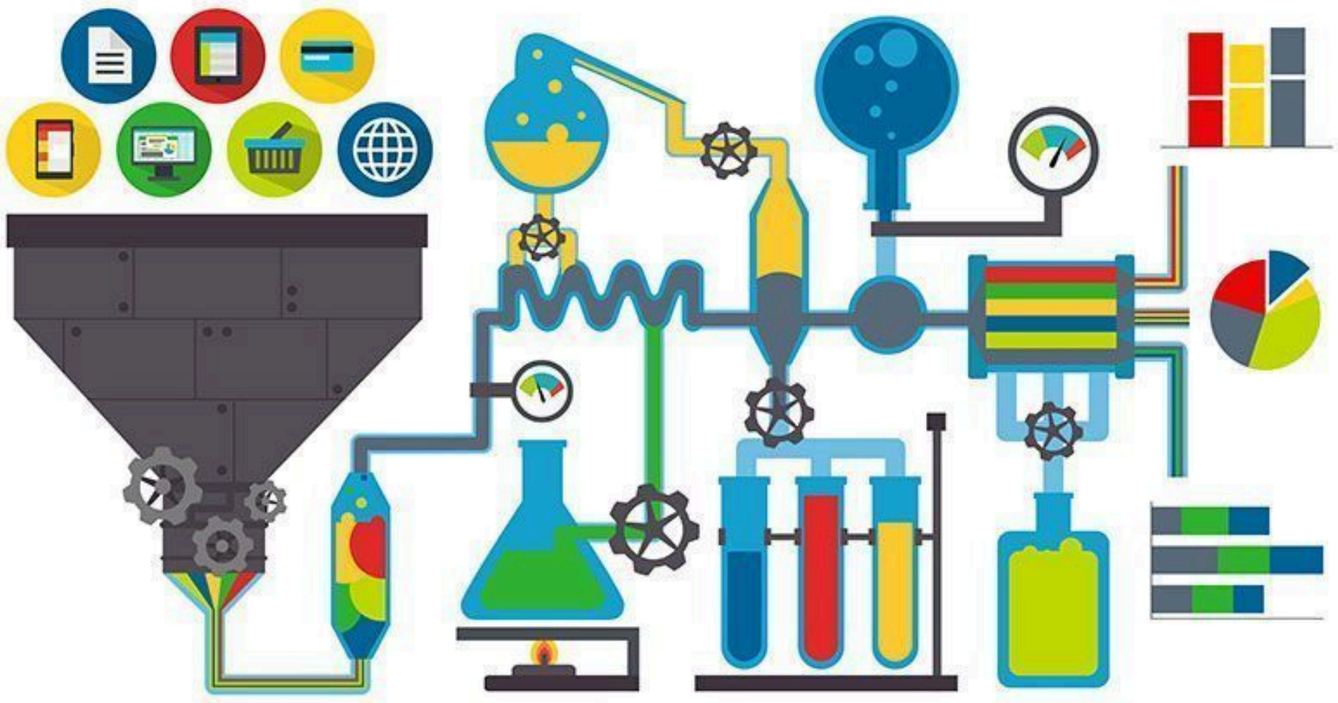
☑ ¿QUÉ ES UN DATO?

¿DATO = INFORMACIÓN?

¿DATO = CONOCIMIENTO?

DATO - INFORMACIÓN - CONOCIMIENTO

Fuente: https://es.calameo.com/books/0044773690d316020e3b5



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- DESCRIPCIÓN DE LA METODOLOGÍA KDD -

KDD: KNOWLEDGE DISCOVERY IN DATABASES

Fayyad, Piatetsky-Shapiro, & Smyth (1996) definen KDD como:

«Knowledge Discovery in Databases is the non trivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data»

Imagen: https://stock.adobe.com/es/images/kdd-knowledge-discovery-in-databases-inscription-on-green-keyboard-key/393044662

METODOLOGÍA KDD

Fayyad, Piatetsky-Shapiro, & Smyth (1996) definen a la Metodoología KDD como:

«The process of using the database along with any required selection, preprocessing, subsampling, and transformations of it; to apply data mining methods (algorithms) to enumerate pat terns from it; and to evaluate the products of data mnining to identify the subset of the enumerated patterns deemed "knowledge".»

Imagen: https://stock.adobe.com/es/images/kdd-knowledge-discovery-in-databases-inscription-on-green-keyboard-key/393044662

METODOLOGÍA KDD

Fayyad (1997) define a la Metodoología KDD como:

«El descubrimiento de conocimiento en bases de datos es un campo de la inteligencia artificial de rápido crecimiento, que combina técnicas del aprendizaje de máquina, reconocimiento de patrones, estadística, bases de datos, y visualización para automáticamente extraer conocimiento (o información), de un nivel bajo de datos (bases de datos)»

Imagen: https://stock.adobe.com/es/images/kdd-knowledge-discovery-in-databases-inscription-on-green-keyboard-key/393044662

METODOLOGÍA KDD

Molina (2001) define a la Metodoología KDD como:

«La extracción no trivial de información potencialmente útil a partir de un gran volumen de datos, en el cual la información está implícita, donde se trata de interpretar grandes cantidades de datos y encontrar relaciones o patrones, para conseguirlo harán falta técnicas de aprendizaje, estadística y bases de datos»

Imagen: https://stock.adobe.com/es/images/kdd-knowledge-discovery-in-databases-inscription-on-green-keyboard-key/393044662

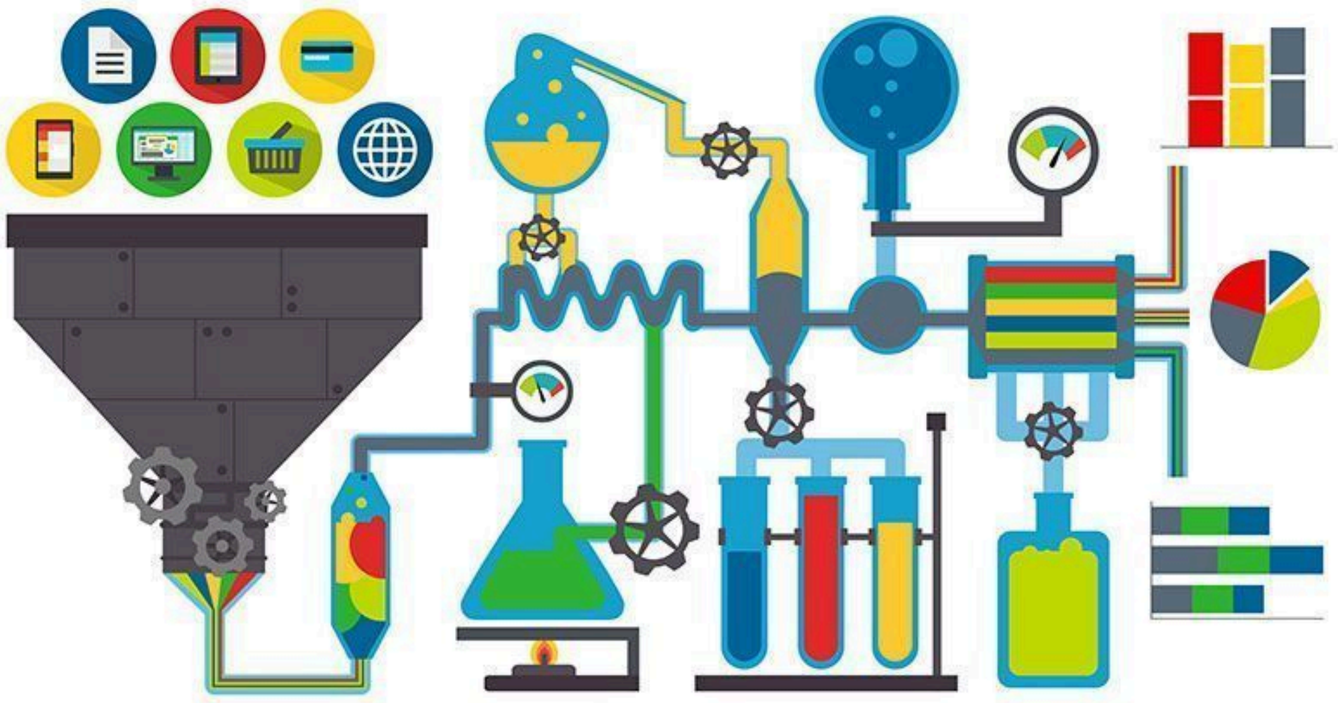


Imagen: http://www.deepphat.co.uk/2018/04/data-miner.html

- ETAPAS DE LA METODOLOGÍA KDD -

METODOLOGÍA KDD

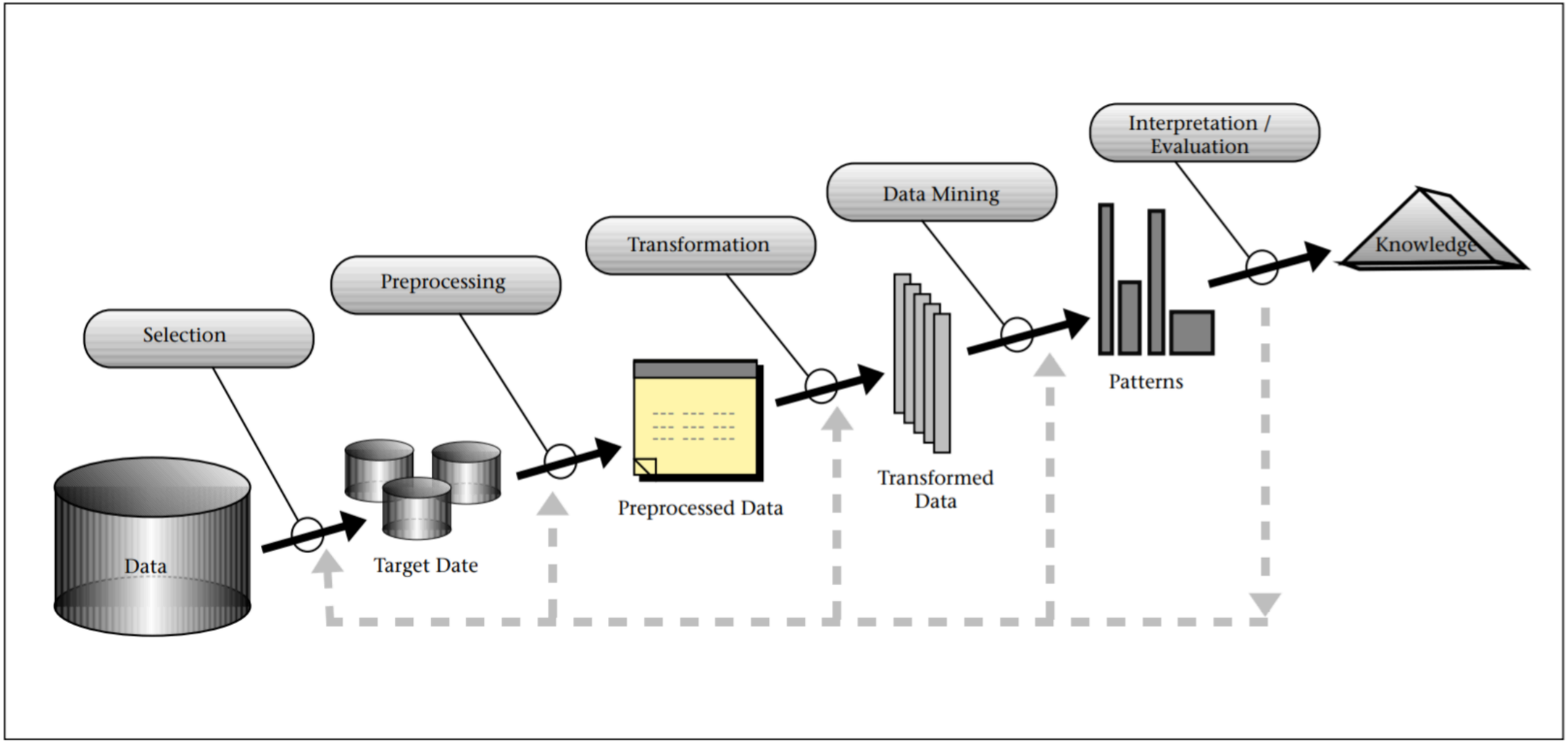


Imagen: "From Data Mining to Knowledge Discovery in Databases". Usama Fayyad, Gregory Piatetsky-Shapiro, and Padhraic Smyth (1996)

9 STEPS

Fuente: Fayyad, Piatetsky-Shapiro, & Smyth (1996)

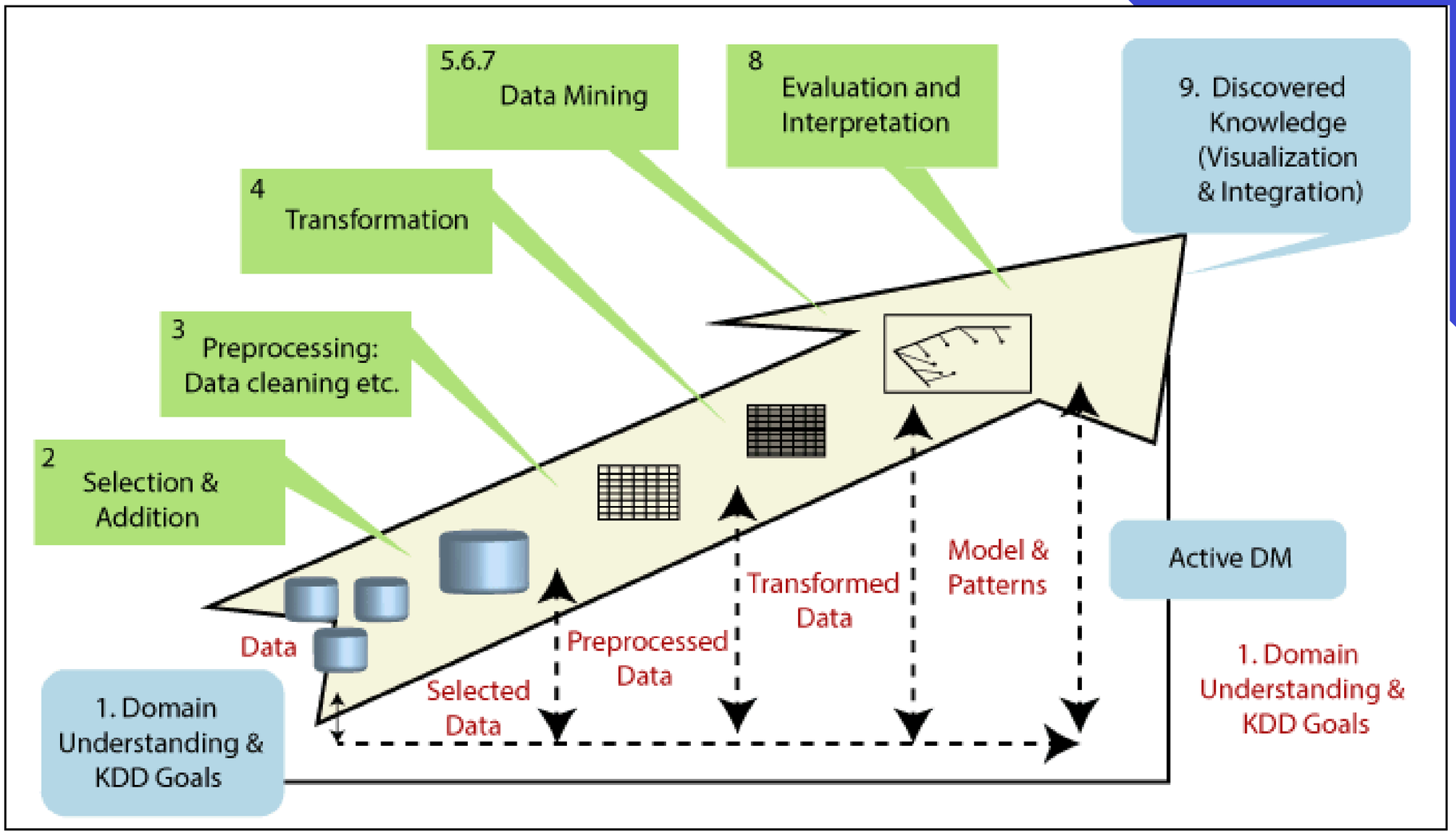
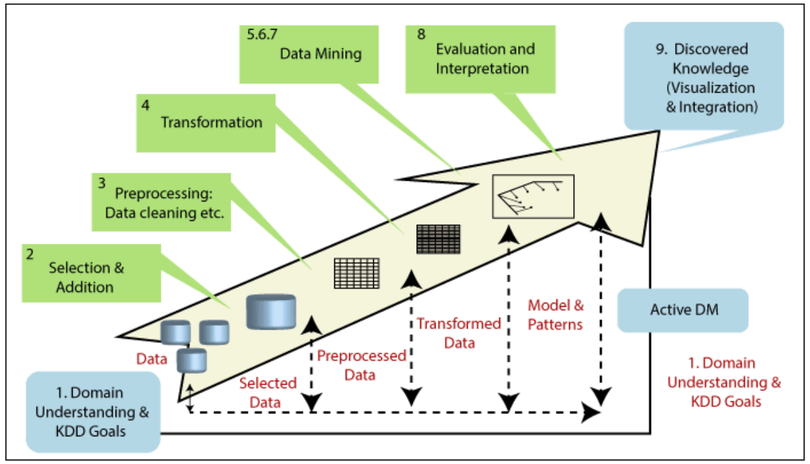
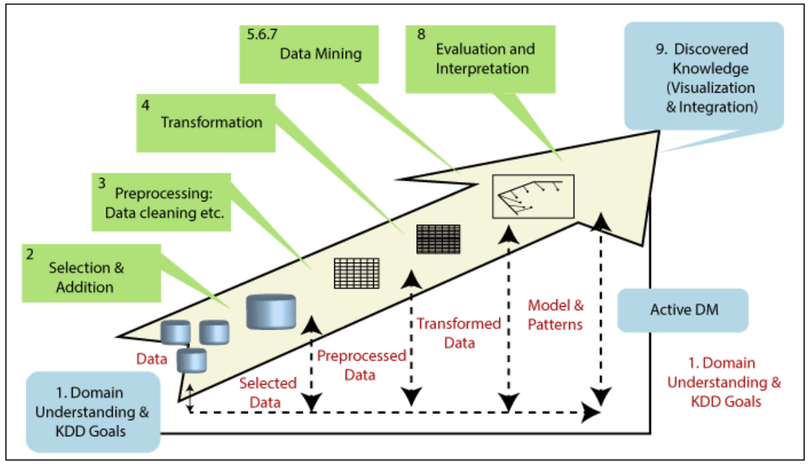


Imagen: https://www.javatpoint.com/kdd-process-in-data-mining



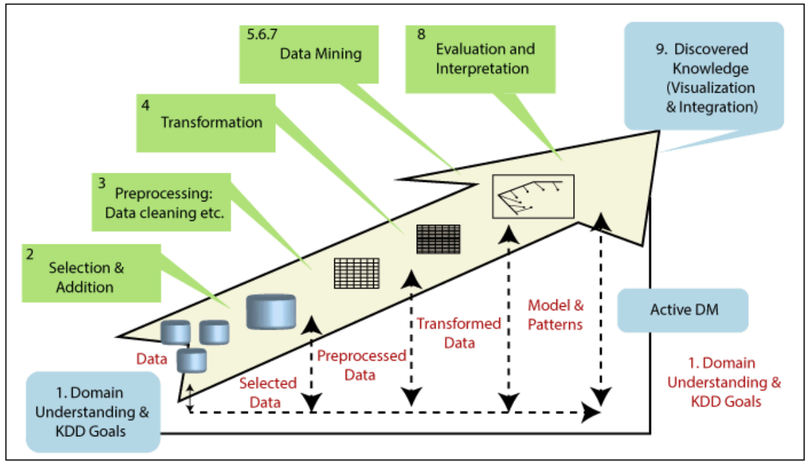
STEP 1

«Developing an **understanding** of the application domain and the relevant prior knowledge, and identifying the goal of the KDD process from the **customer's viewpoint**.»



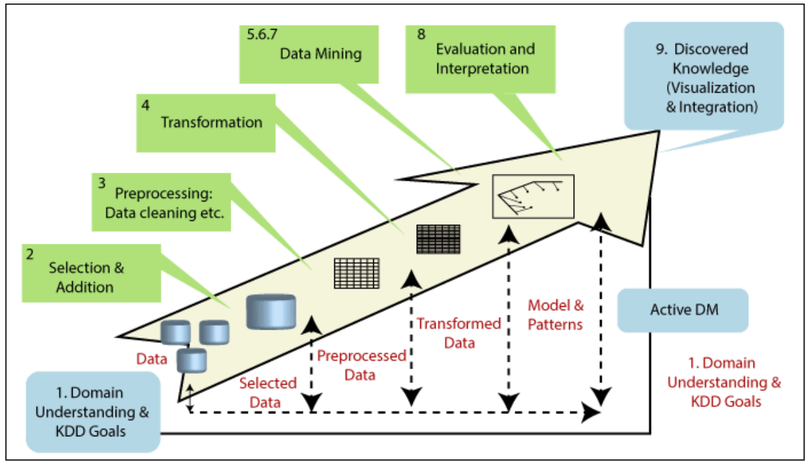
STEP 2

«**Creating a target data set**: selecting a data set, or focusing on a subset of variables or data samples, on which discovery is to be performed.»



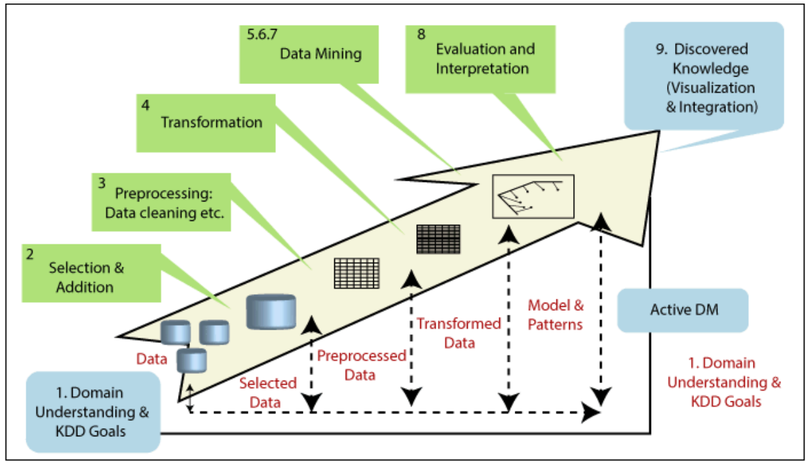
STEP 3

«Data cleaning and preprocessing: basic operations such as the removal of noise and strategies for handling missing data fields.»



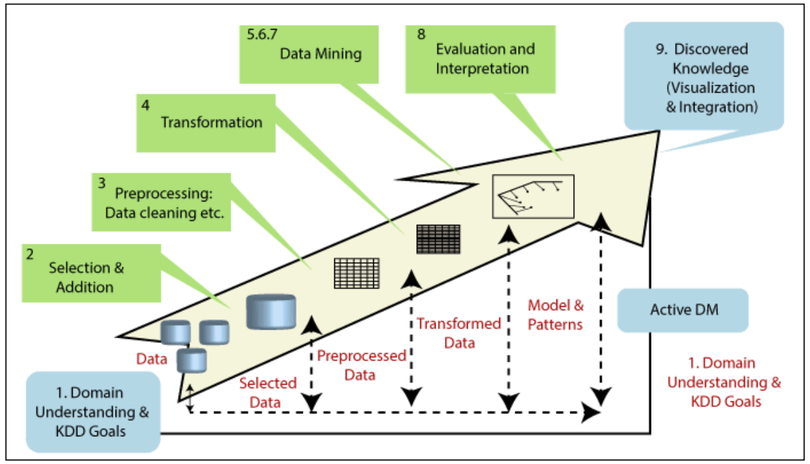
STEP 4

«**Data reduction and projection**: finding useful features to represent the data. Using methods to reduce the number of variables.»



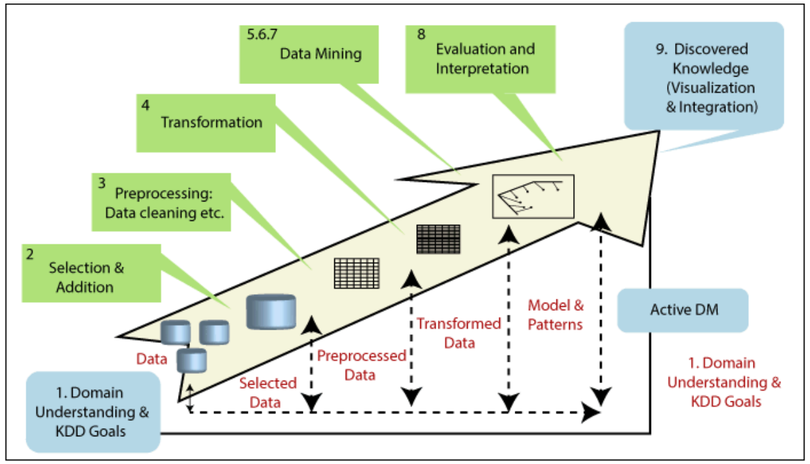
STEP 5

«Matching the goals of the KDD process to particular **data mining method**: summarization, classification, regression, clustering, etc.»



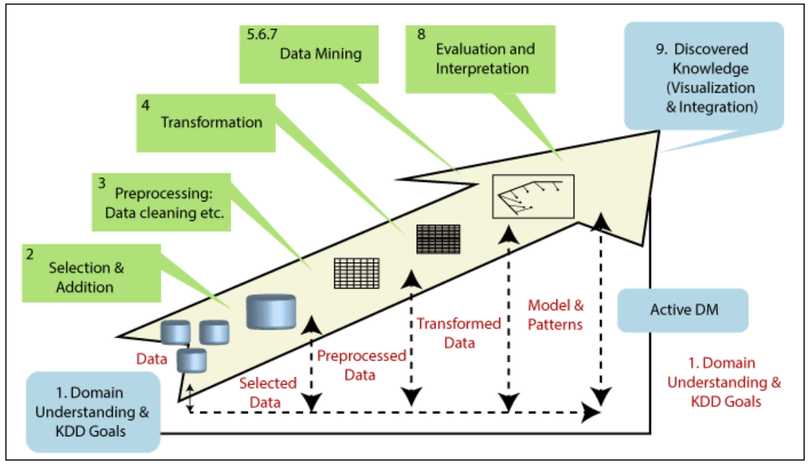
STEP 6

«**Choosing the data mining algorithms**: which models and parameters may be appropriate for searching for patterns (e.g. models for categorical data or models on vectors over the reals).»



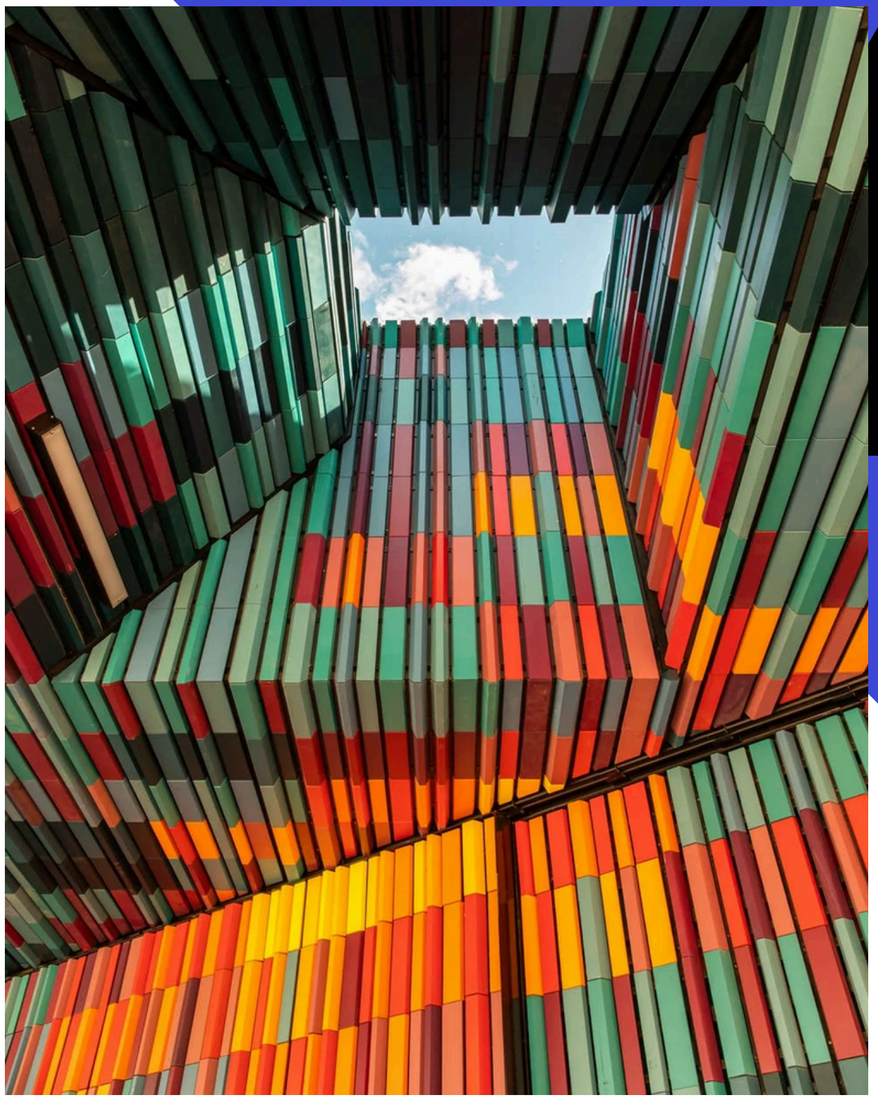
STEP 7

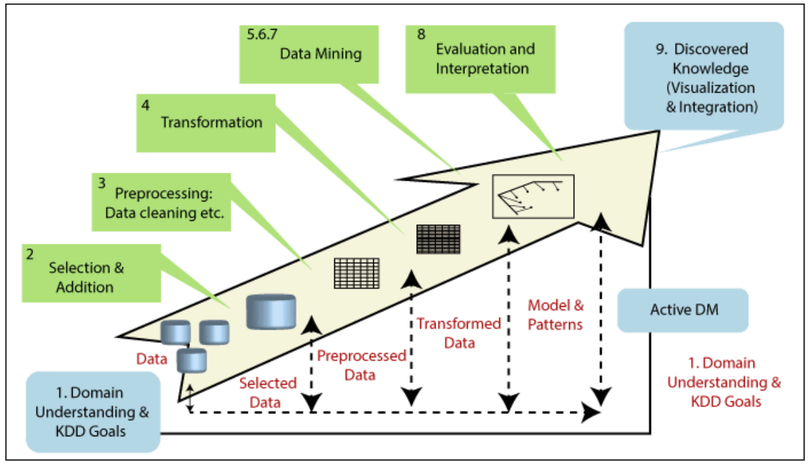
«**Data mining**: searching for patterns of interest: classification trees, regression, clustering, and so forth. The user can significantly aid by correctly performing the preceding steps.»



STEP 8

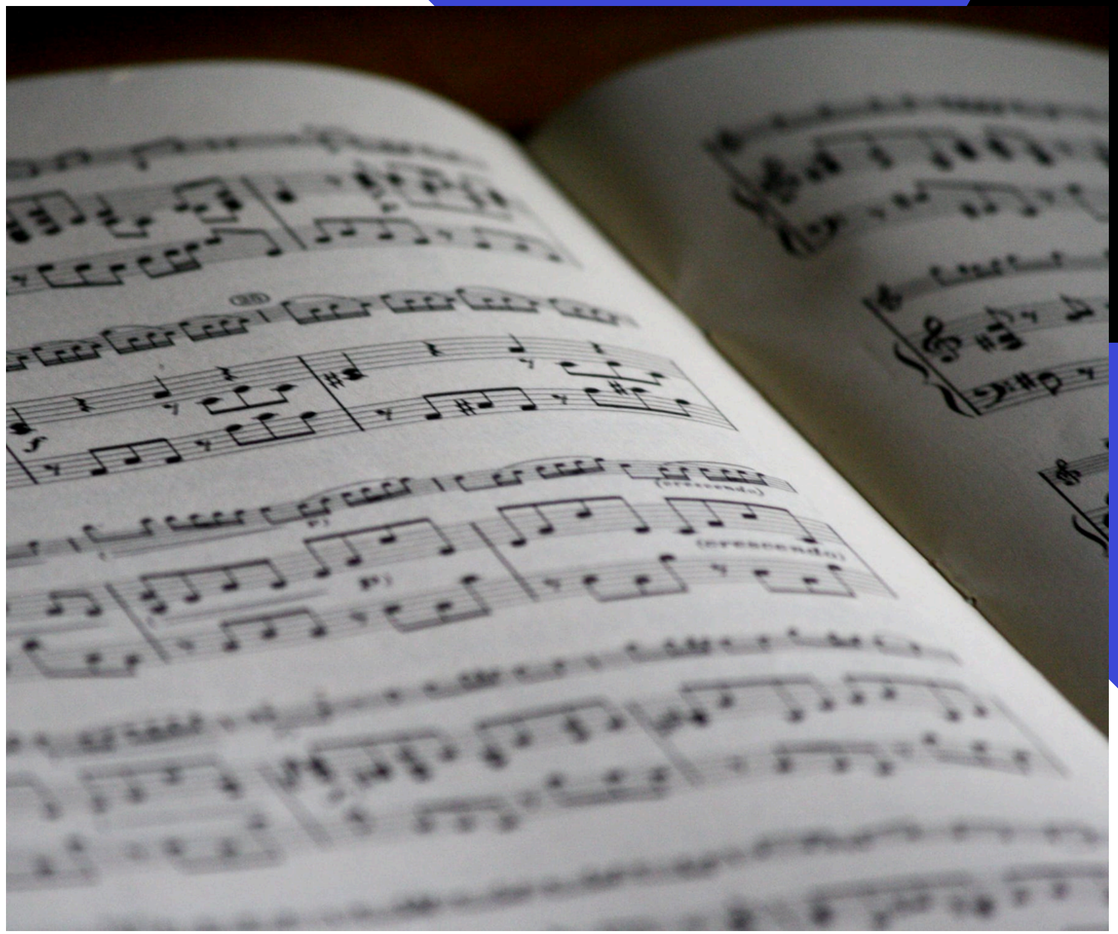
«**Interpreting mined patterns**: possibly return to any steps. Also involve visualization of the extracted patterns/models, or visualization of the data given the extracted models.»





STEP 9

«**Consolidating discovered knowledge**: incorporating this knowledge into another system for further actions, or documenting it and reporting it to interested parties.»



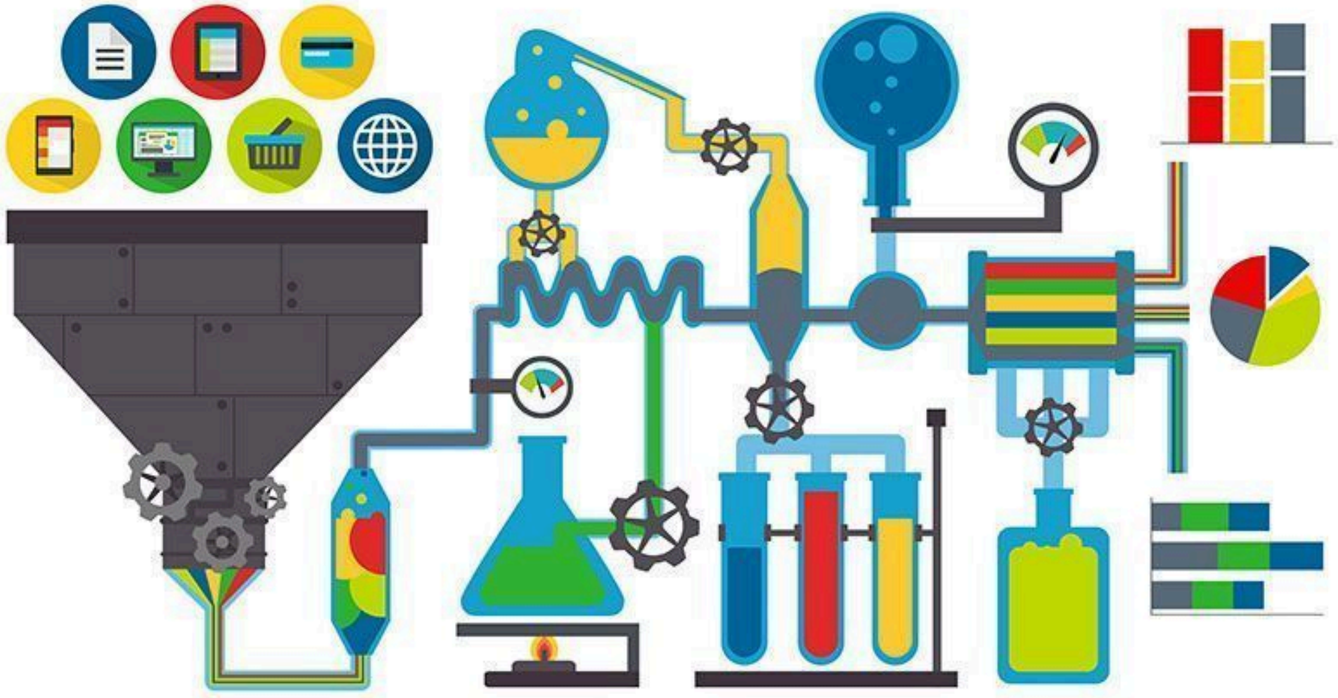


Imagen: http://www.deepphat.co.uk/2018/04/data-miner.html

- THE DATA MINING STEP OF THE KDD PROCESS -

DATA MINING METHODS

«IN THE CONTEXT OF KDD, **DESCRIPTION** TENDS TO BE MORE IMPORTANT THAN **PREDICTION**. THIS IS IN CONTRAST TO MANY MACHINE LEARNING AND PATTERN RECOGNITION APPLICATIONS WHERE **PREDICTION** IS OFTEN THE **PRIMARY GOAL»**

Fuente: Fayyad, Piatetsky-Shapiro, & Smyth (1996)

DATA MINING METHODS

THE GOALS OF PREDICTION AND DESCRIPTION ARE ACHIEVED VIA THE FOLLOWING PRIMARY DATA MINING METHODS

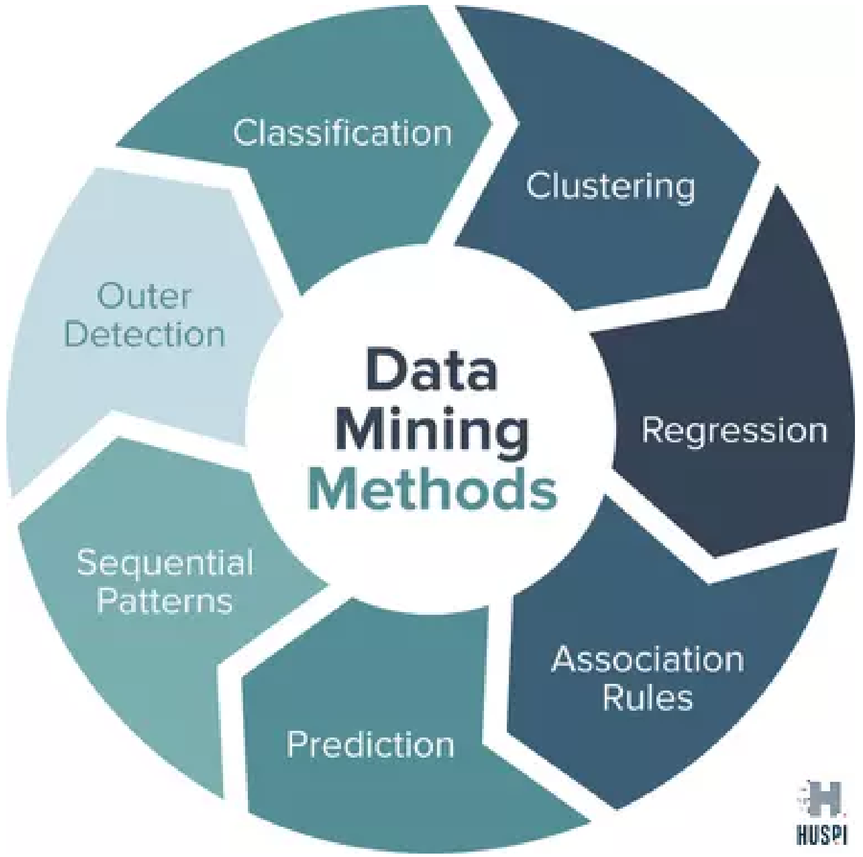


Imagen: https://huspi.com/blog-open/what-is-data-mining/

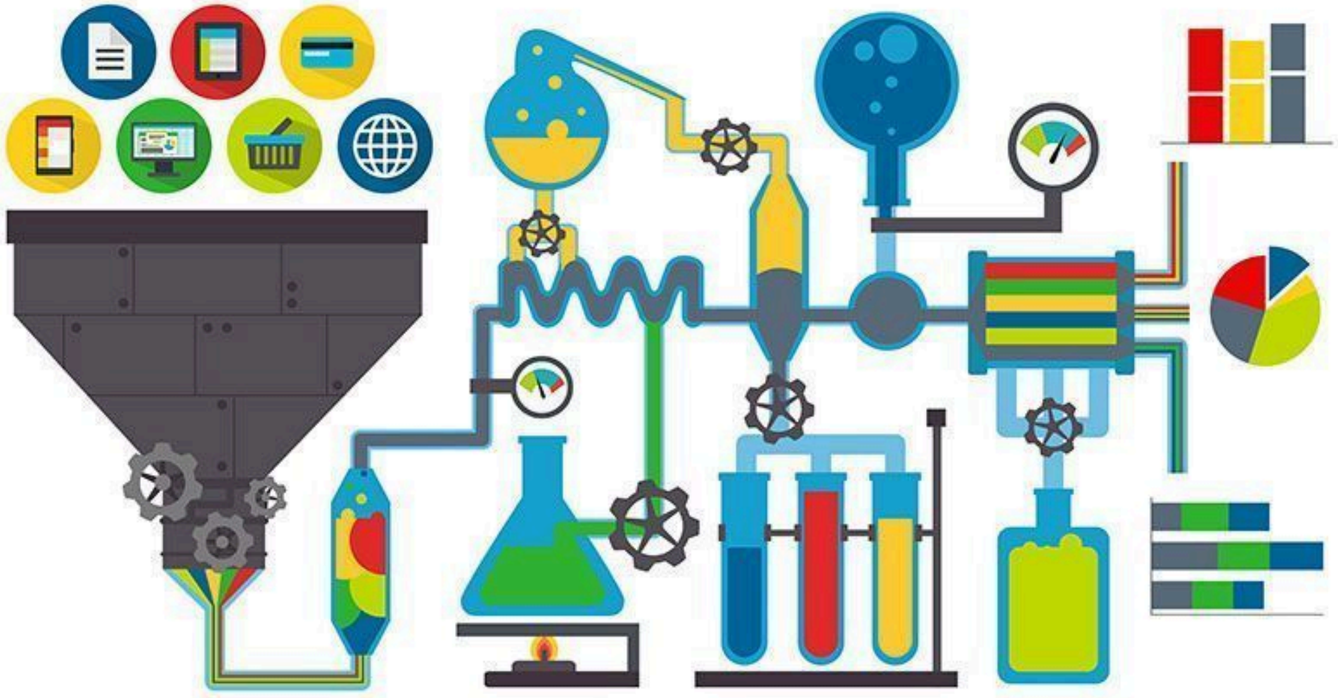


Imagen: http://www.deepphat.co.uk/2018/04/data-miner.html

- OTRAS METODOLOGÍAS -

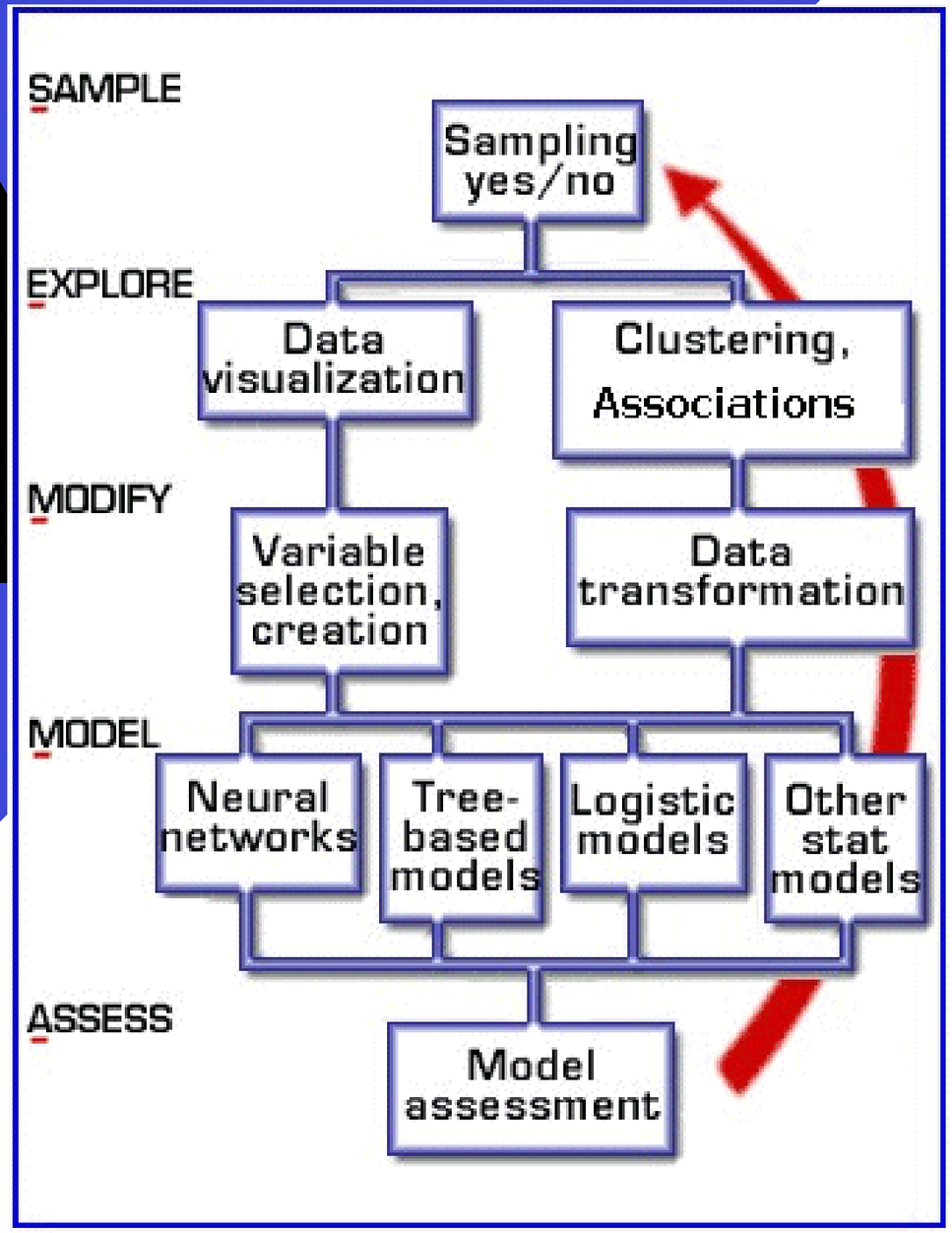
Fuente: https://healthdataminer.com/data-mining/crisp-dm-una-metodologia-para-mineria-de-datos-en-salud/

CRISP-DM

"Cross Industry Standard Process for Data Mining"

Fases propuestas:

1- Comprensión del negocio
2- Compresión de los datos
3- Preparación de los datos
4- Modelado
5- Evaluación
6- Despliegue



Fuente: https://documentation.sas.com/doc/en/emref/14.3/n061bzurmej4j3n1jnj8bbjjm1a2.htm

SEMMA

SAS Institute defines data mining as the process of:

- **Sampling,**
- **Exploring,**
- **Modifying,**
- **Modeling, and**
- **Assessing**

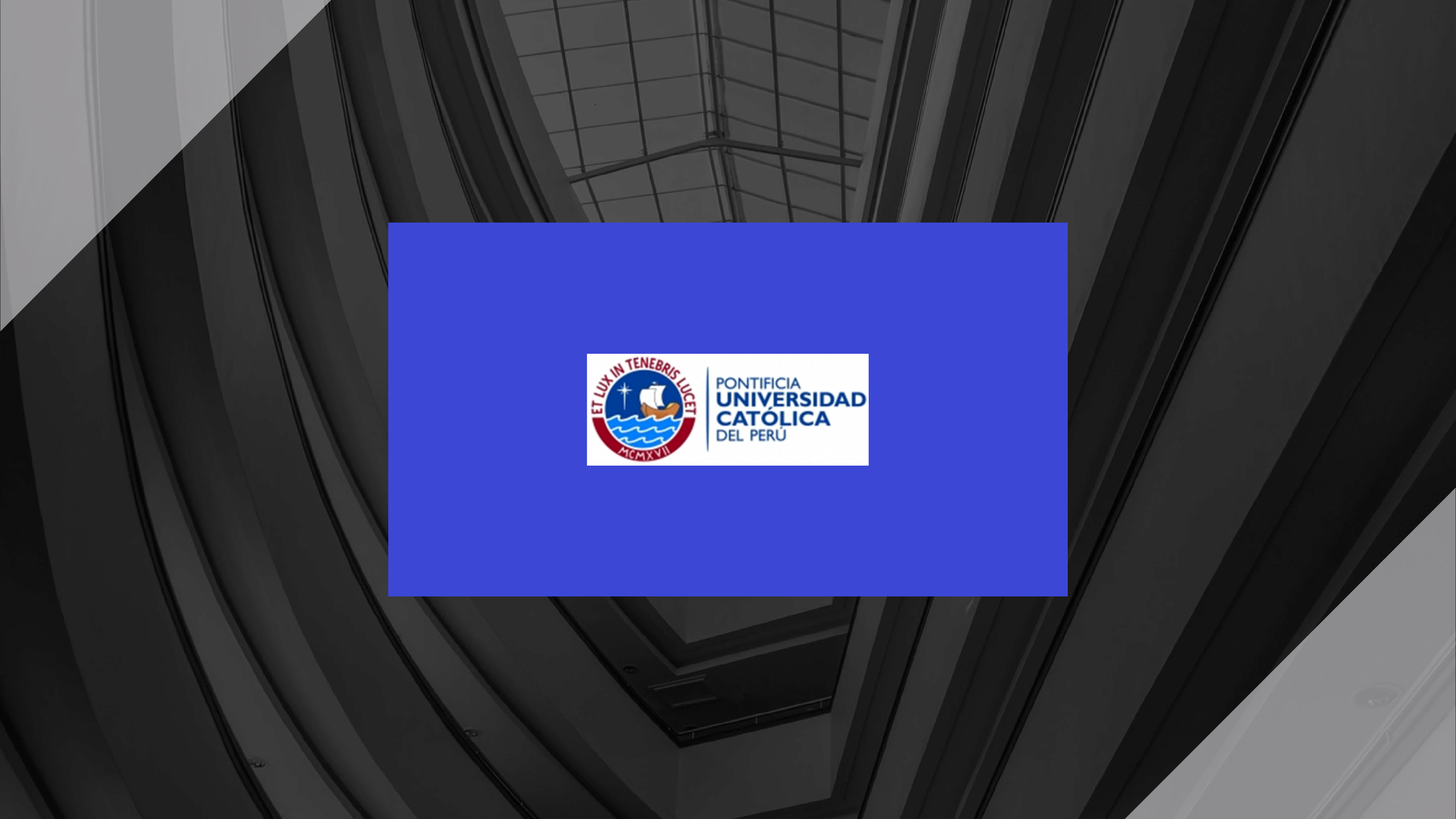
large amounts of data to uncover previously unknown patterns which can be utilized as a business advantage.

Comparativo de Metodologías

Data Mining Process Models	KDD	CRISP-DM	SEMMA
No. of Steps	9	6	5
Name of Steps	Developing and Understanding of the Application	Business Understanding	---------
	Creating a Target Data Set	Data Understanding	Sample
	Data Cleaning and Pre-processing		Explore
	Data Transformation	Data Preparation	Modify
	Choosing the suitable Data Mining Task	Modeling	Model
	Choosing the suitable Data Mining Algorithm		
	Employing Data Mining Algorithm		
	Interpreting Mined Patterns	Evaluation	Assessment
	Using Discovered Knowledge	Deployment	---------

Imagen: https://medium.com/analytics-vidhya/knowledge-discovery-data-kdd-a8b41509bff9

LENGUAJES DE PROGRAMACIÓN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